
Transfection

Introduction

Creating retrovirus for a plasmid.

Materials

› Material

›

Procedure

Procedure

Day 1: Plating Cells

1. **Use medium without PenStrep** (improves transfection efficiency)

2. **Plate cells as follows:**

10 cm dish:

2.5×10^6 cells

10 mL medium

6-well plate:

0.5×10^6 cells per well

2 mL medium per well

Day 2: Transfection

3. **Remove media**, replace with **fresh media from Plate E cells**

4. Prepare **transfection mix (per well):**

Tube 1:

5 μ g (= 50 μ L) pCL-Eco plasmid

550 μ L OptiMEM

Aliquot 600 μ L into 15 mL Falcon tubes

Add 10 μ g of transfer plasmid

Tube 2:

45 μ L Mirus transfection reagent

655 μ L OptiMEM

Combine tubes:

Add 700 μ L of Tube 2 mix into each Falcon tube

Incubate at **room temperature for 20 min**

5. **Add dropwise to cells**

Day 3

6. Remove old media

7. Add **fresh TCM media**

Day 4

8. Harvest **first round of virus (supernatant)**

9. Store in fridge

10. Add **fresh TCM media** to cells

Day 5

11. Harvest **remaining virus**, combine with Day 4 supernatant

12. Spin at **1600 rpm for 4 min**

13. Aliquot:

2 mL per 5 mL Eppi or use directly

14. **Freeze at -80°C**

15. If using **fresh virus**, filter through **0.45 μ m filter**